UNIVERSITY OF CAMBRIDGE | Department of Engineering

Manufacturing CMOS Integrated Circuits

Fabrication, design rules and layout

Suggested reading: Hodges 3.1-3.3; Rabaey 2

Electrical Engineering CAMBRIDGE

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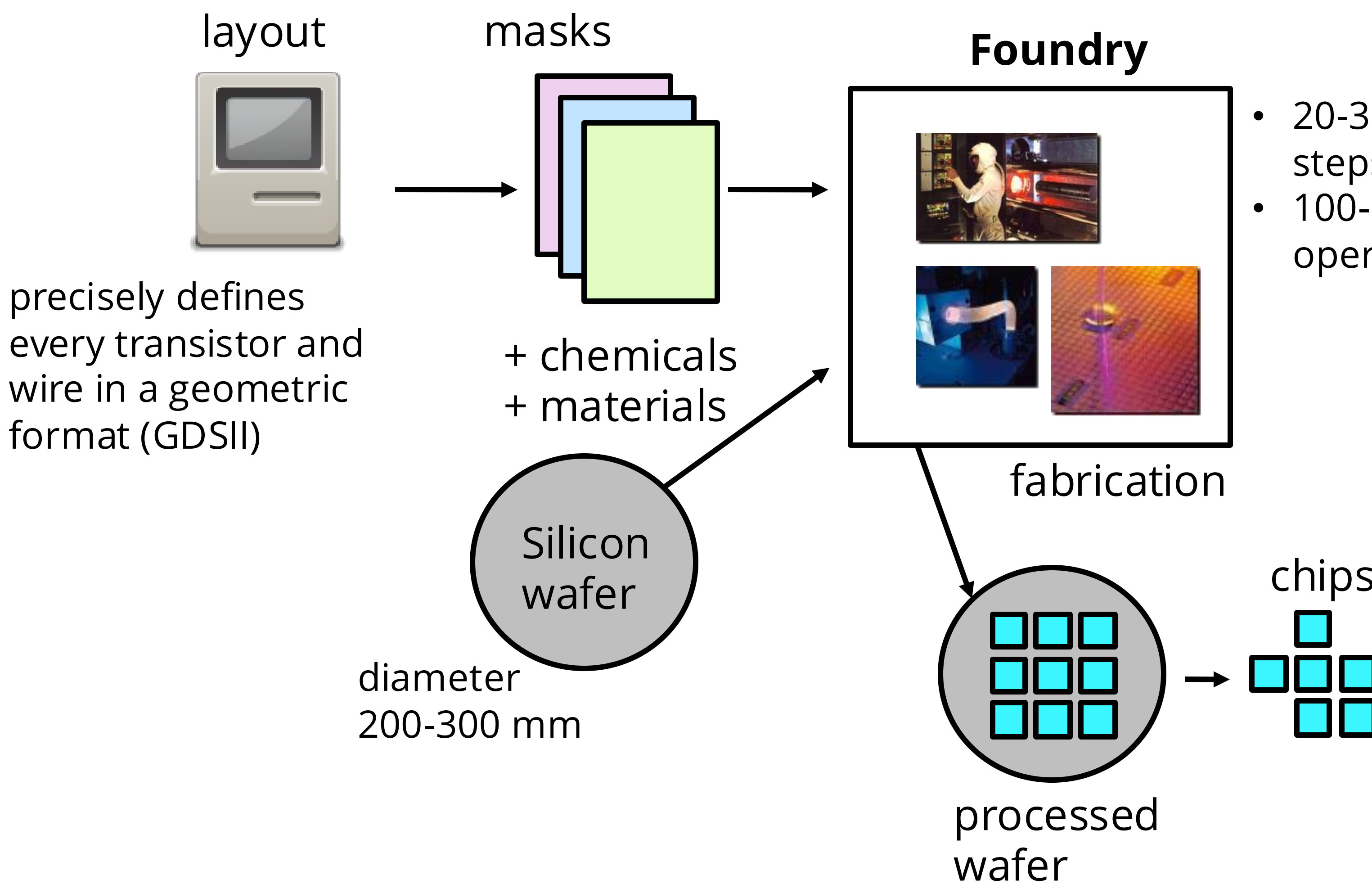
Learning Outcomes

- To gain insight into the steps and issues associated with the manufacture of integrated circuits
- To understand the role of design rules to bridge the IC design and the complex manufacturing process
- To practise the skill of projecting a planar IC layout into a three-dimensional layering structures created during fabrication

Introduction

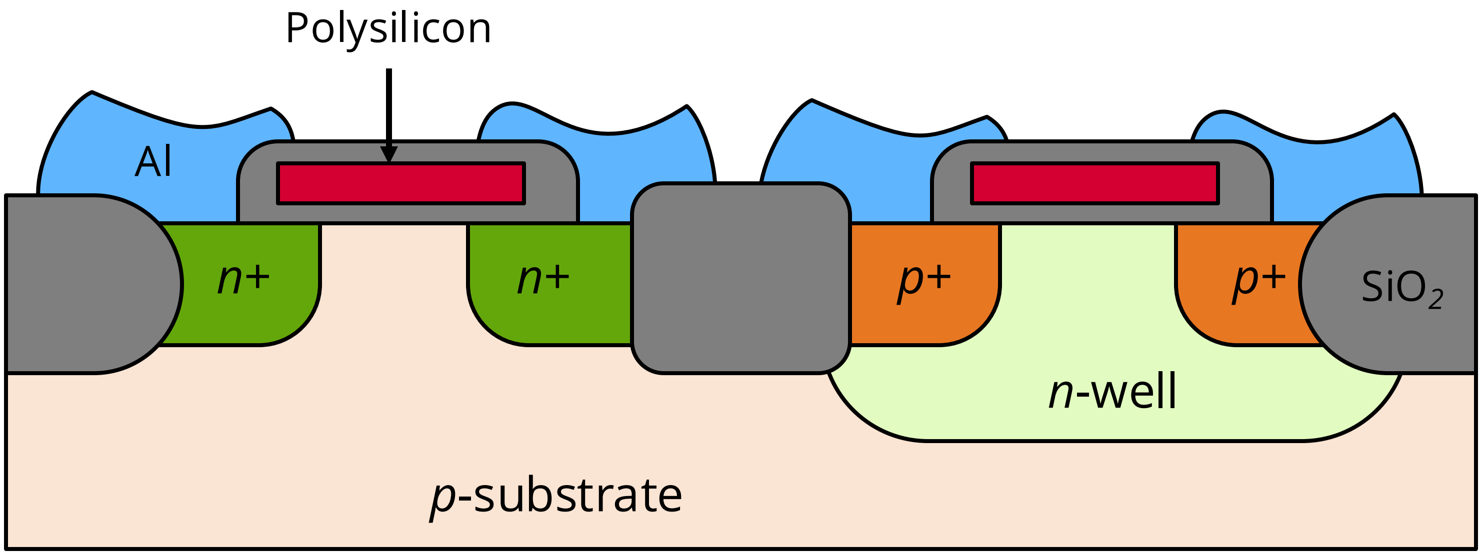
- Some insights into the steps that lead to an operational silicon chip comes in handy for chip designers.
- A detailed description of the fabrication technology deserves a complete course.
- We are going to briefly describe the steps and techniques used in a modern integrated circuit manufacturing process.
- The fabrication process has never been static – constant innovations keep Moore's law going.

The Chip Making Process



- 20-30 major steps
- 100-200 distinct operations

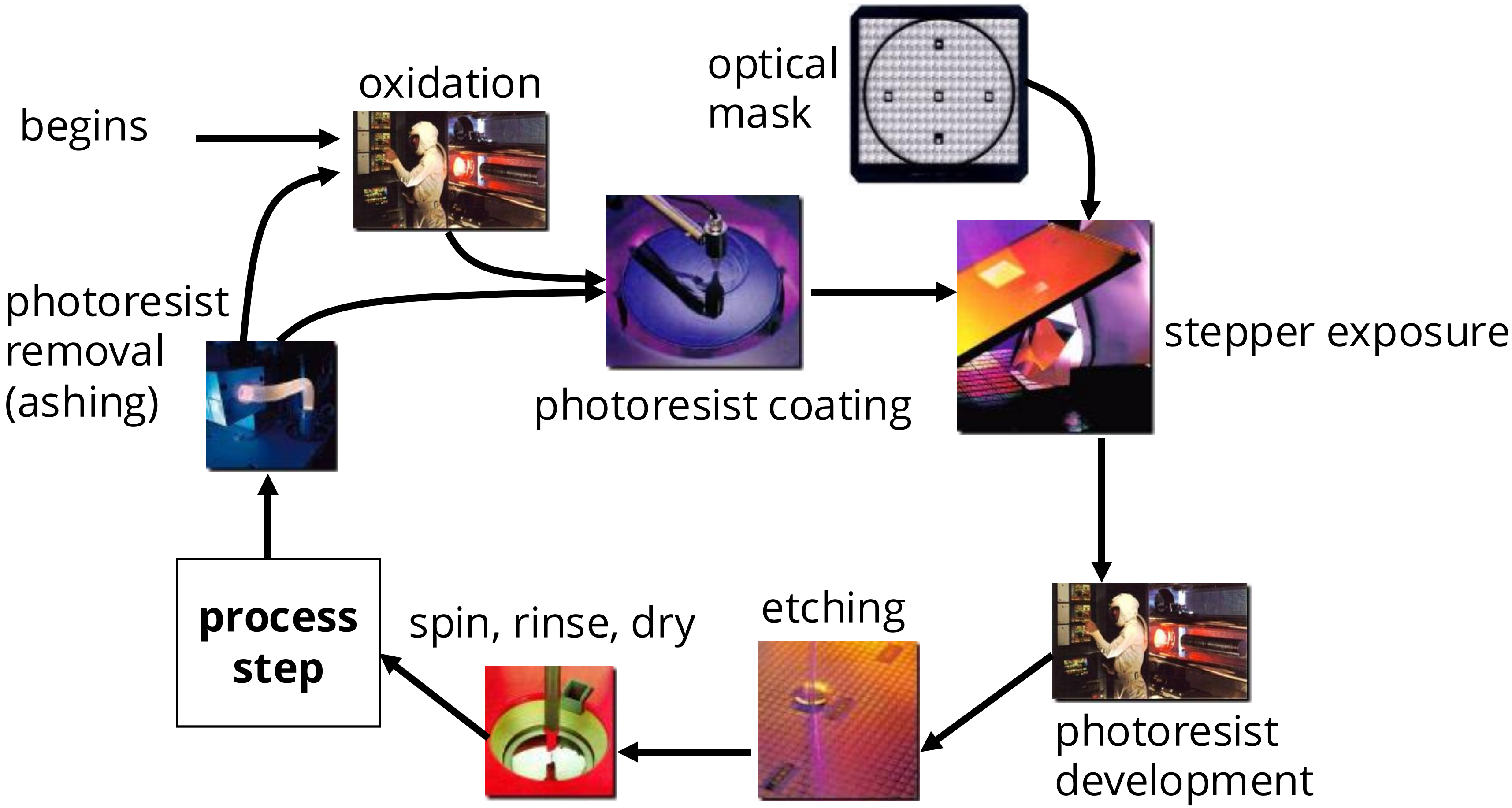
Traditional CMOS Process



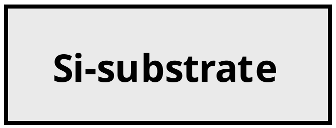
- *n*-well CMOS process:
  NMOS implemented in *p*-doped substrate and PMOS inside *n*-well.
- Aluminum (Al) used for creating contacts to sources and drains
- Silicon oxide (SiO_2) is the main insulator to separate (semi)conducting materials

Photolithographic Process

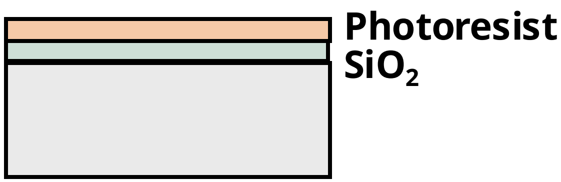
Typical operations in a single photolithographic cycle



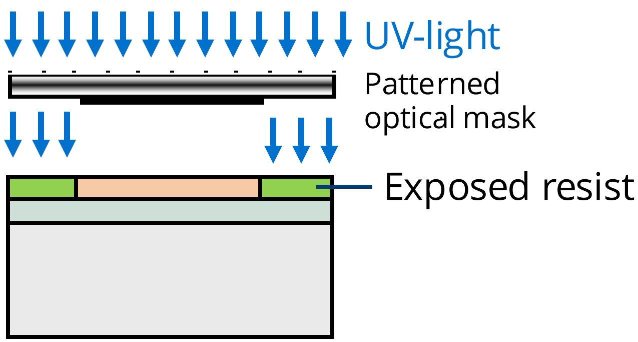
Photolithographic Process (Cont')



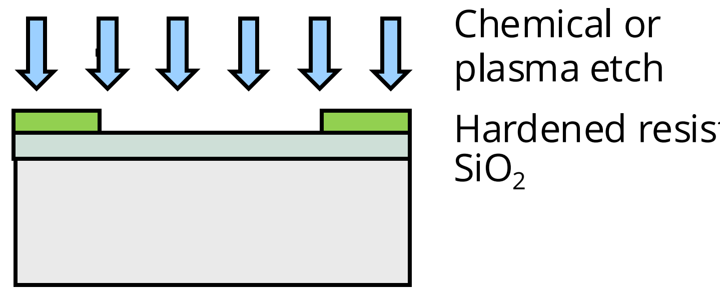
Silicon base material



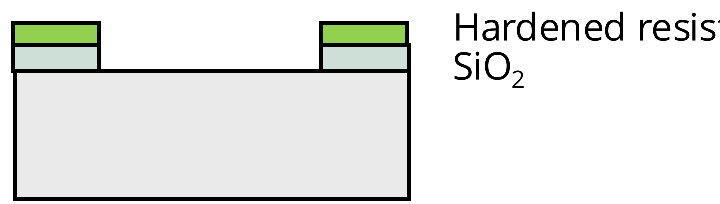
After oxidation and deposition of negative photoresist



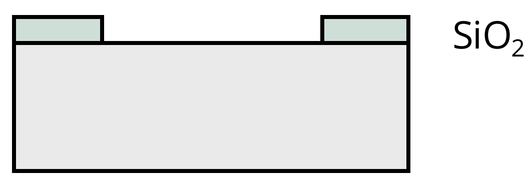
Stepper exposure



After development and etching of resist, chemical or plasma etch of SiO_2.



After etching

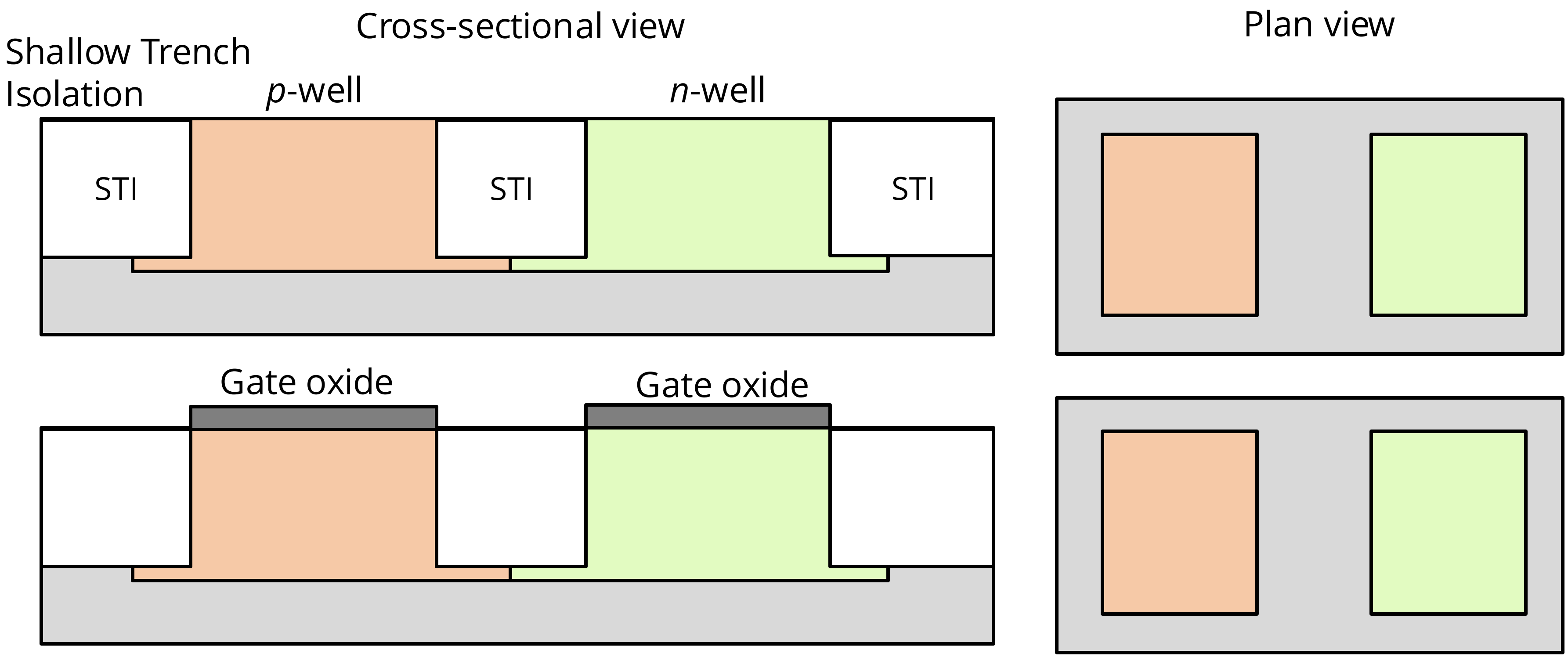


Final result after removal of resist

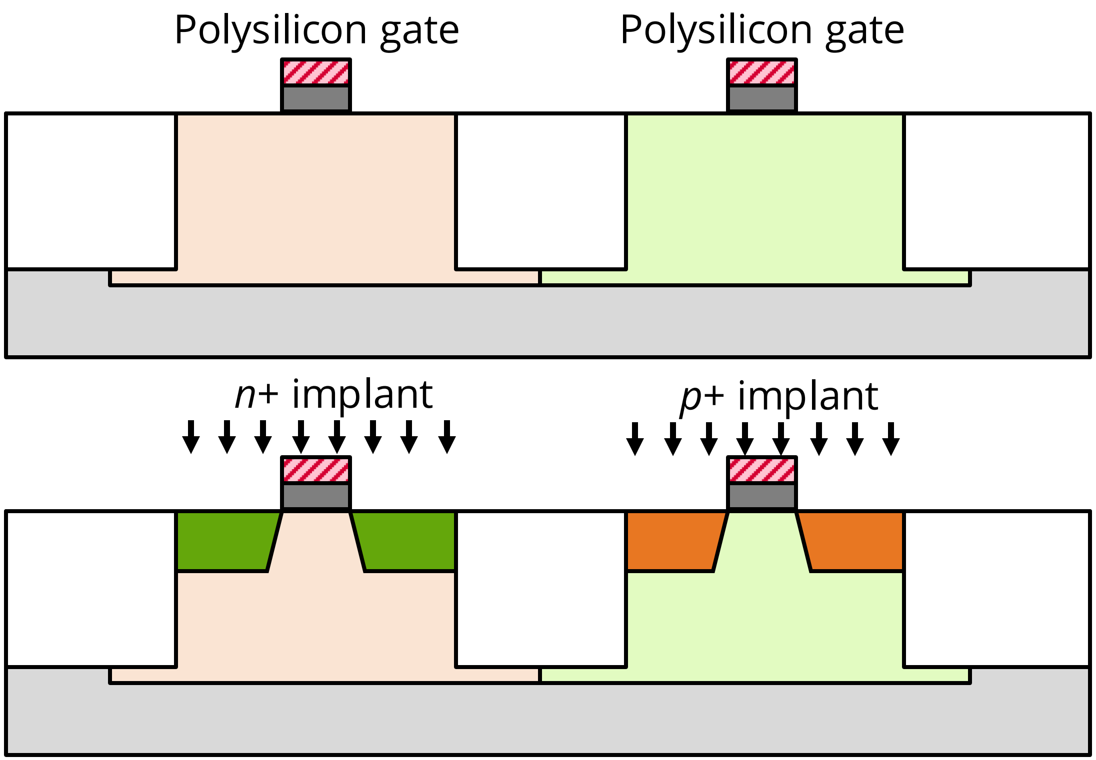
Process Steps

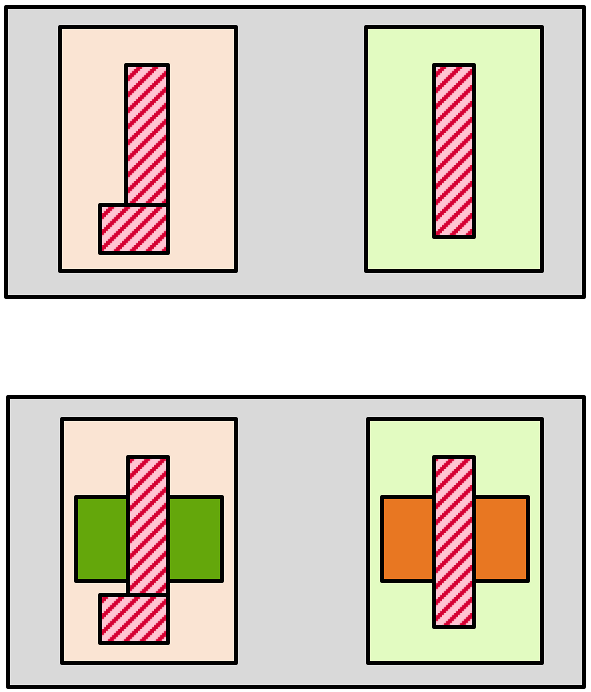
- Diffusion: doping process to form n-type or p-type material by high temperature exposure (~ 900-1100^{o}C) to donor or acceptor impurities
- Ion implementation: high-energy bombardment of silicon with donor or acceptor ions from particle accelerators followed by an annealing step (1,000^{o}C then cool down) to activate implants
- Chemical Vapour Deposition (CVD): materials such as metal or oxide are deposited out of a gaseous mixture.
  - Metals can also be deposited using sputtering.
- Planarization: a chemical-mechanical planarization (CMP) step is used to make the surface approximately flat for the next step

The Twin-Well CMOS Process



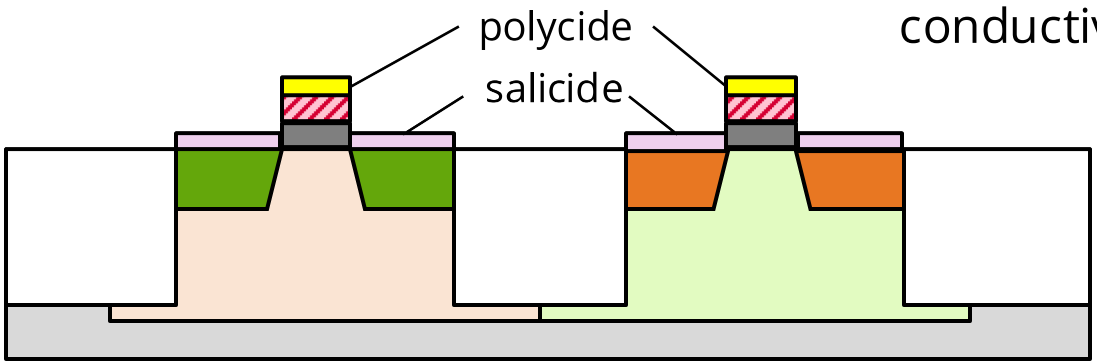
Creating Gates and Diffusions

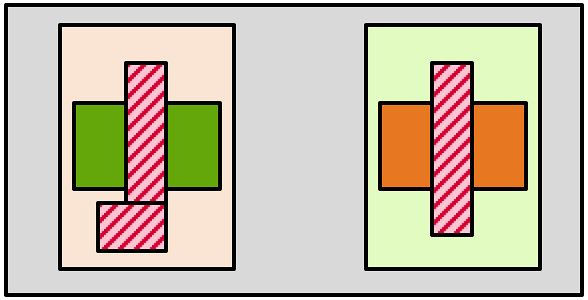


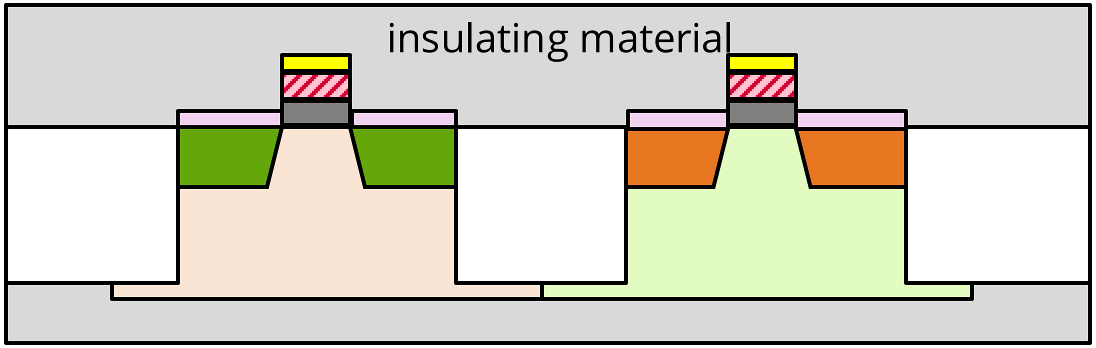


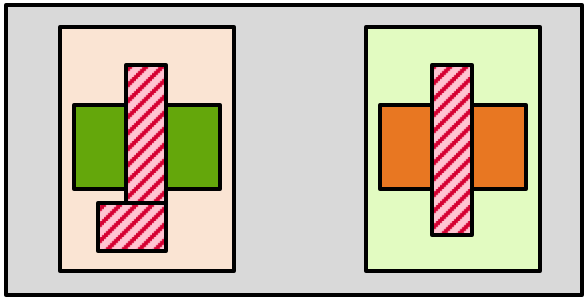
Depositing Silicide Materials

Polycide and salicide are lower resistance compounds to improve conductivity and speed up switches.



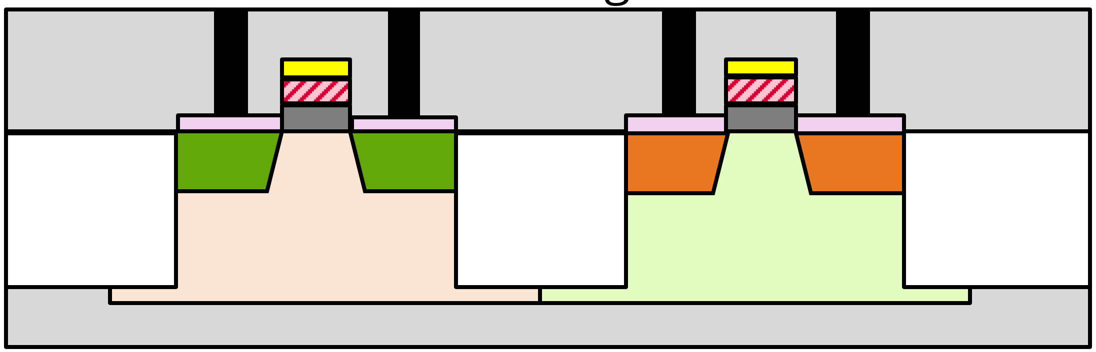


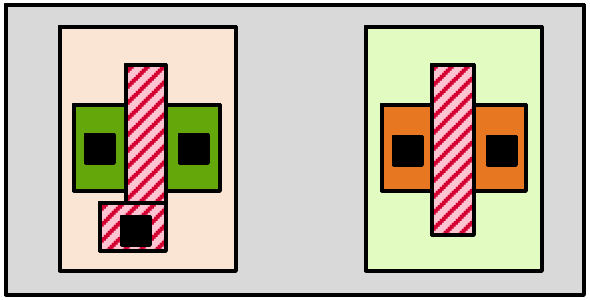




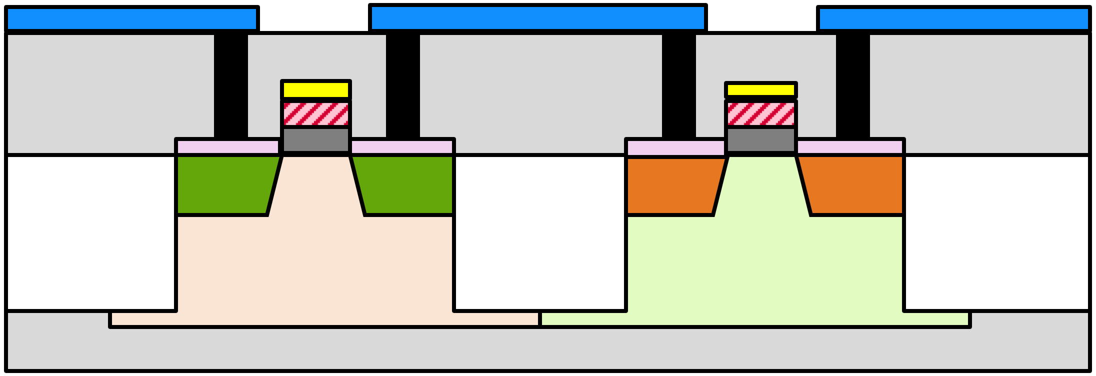
Making Interconnects

Contact cuts filled with tungsten

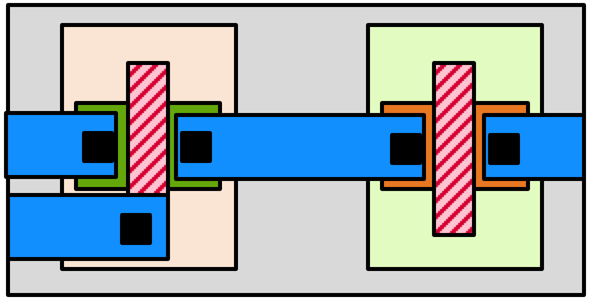




Deposited and patterned metal 1



Copper (Cu) gradually replaces Aluminum (Al) due to better conductivity and durability.



Design Rules

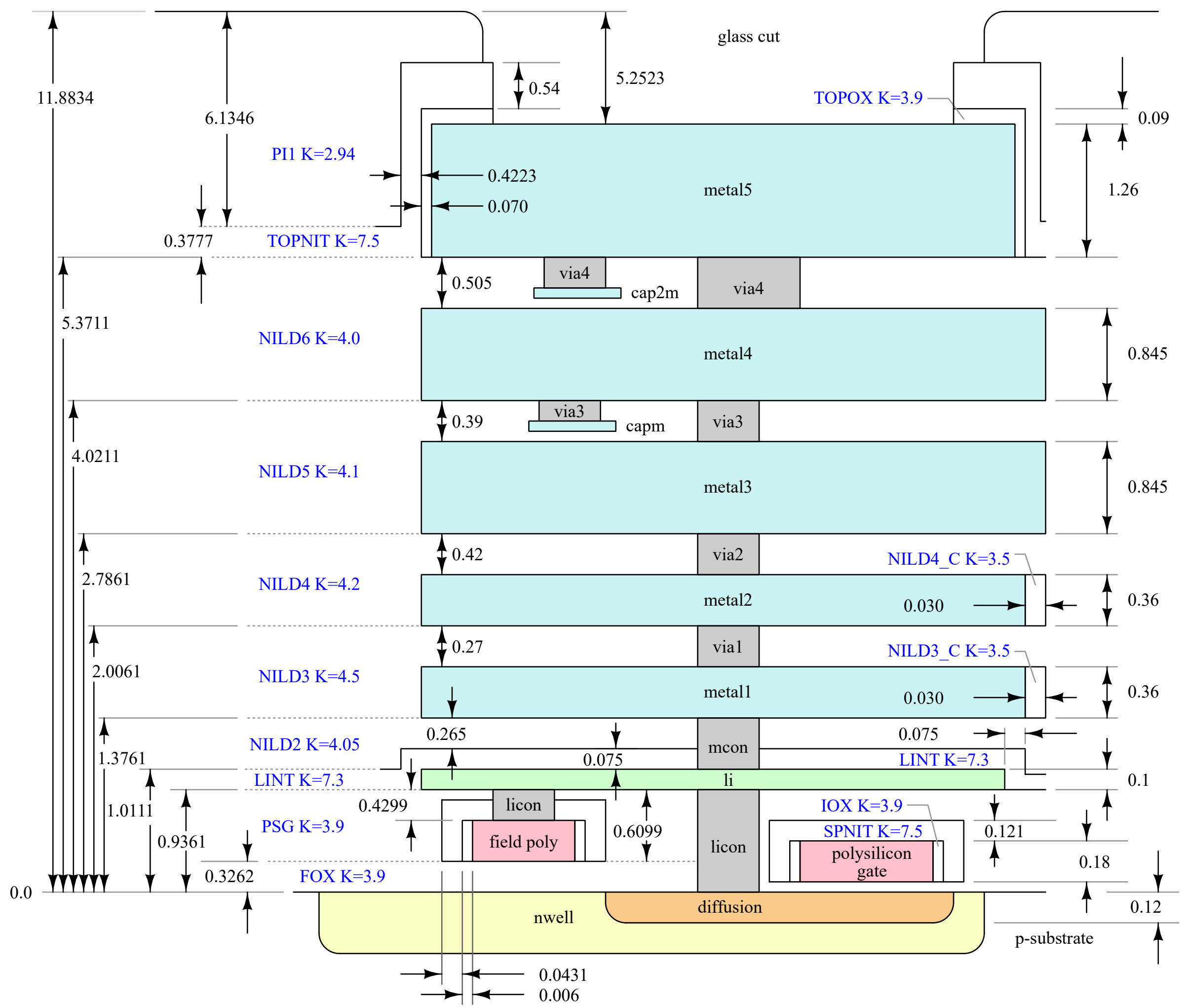
- Goal: allow for a ready translation of a circuit layout into an actual geometry in silicon.
- Rules are enforced by an EDA tool called design rule checkers (DRC).
- Interface/contract between designers and process engineers

Tighter, smaller designs High performance High density	reproducible, high-yield process constraints: min-width, min-spacing
Design house: systems/circuits designers	Foundry (fab): process engineers

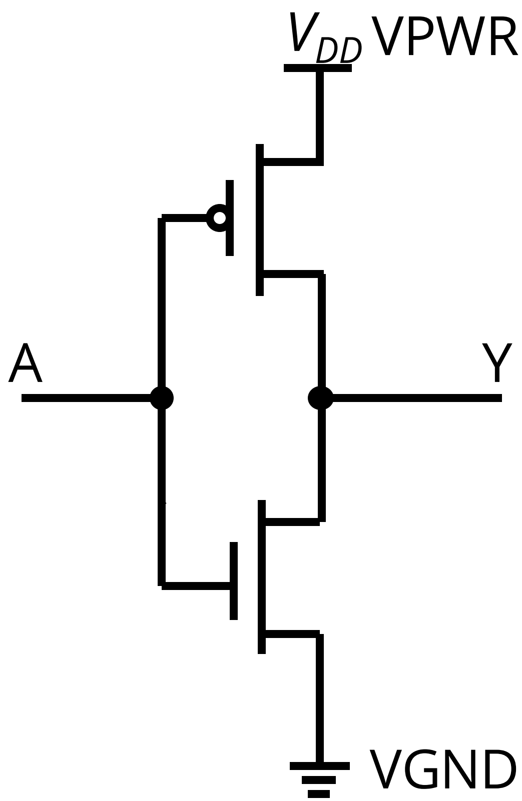
SkyWater SKY130 PDK Process Layers (Reference)

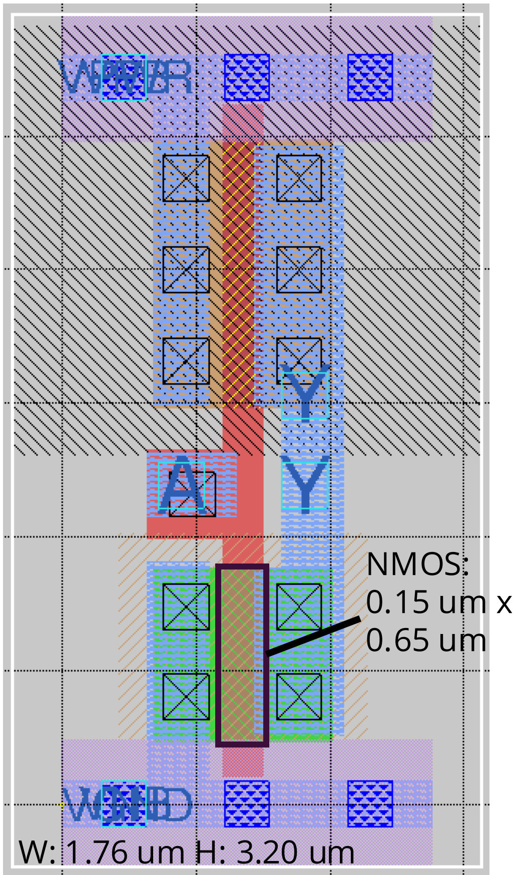
https://skywater-pdk.rtfd.io/

(Diagram not to scale!)

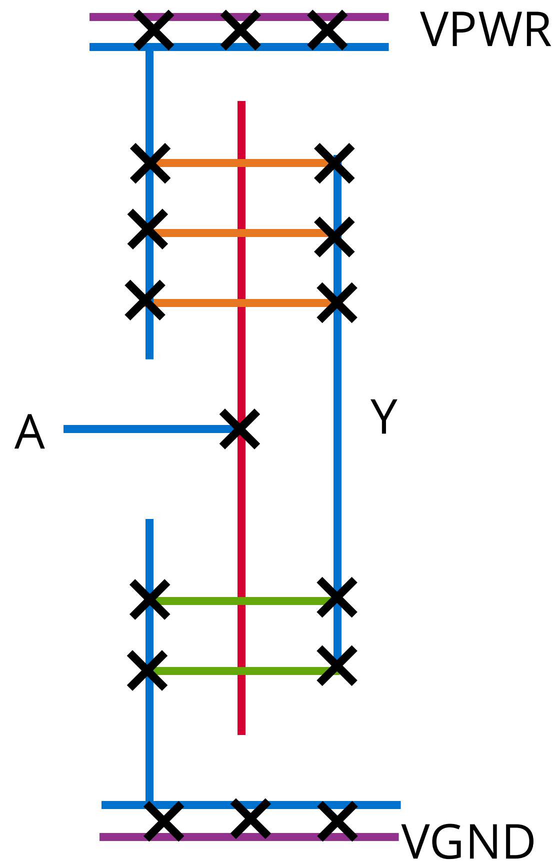


CMOS Inverter: Schematic vs Layout





stick diagram



Summary

- The IC manufacturing process requires many steps, each of which consists of a sequence of operations – photolithographical exposure and development, material deposition and etching.
- The set of design rules defines the constraints in term of minimum width and separation that the IC design has to adhere to if the resulting circuit is to be fully functional after manufacturing.
- It is vital to recognise the layering structure of the integrated circuits during layout stage to make sure of the devices (transistors, capacitors) and interconnects (wires).